Qns	Answer	Marks
4(a)(i)	Power received $= \frac{\text{Area of source}}{4\pi r^2} \times \text{Power of source}$ $= \frac{450 \times 10^{-4}}{4\pi (3500)^2} \times 15.0$ [M1] $= 4.4 \times 10^{-9}$ W [A1]	
4(a)(ii)	Energy from the source radiates uniformly in all directions	B1

Qns	Answer	Marks
4(a)(iii)	Intensity $\propto$ Amplitude^2 Intensity = $\frac{\text{Power}}{4\pi r^2} \propto \frac{1}{r^2}$ Hence $A \propto \frac{1}{r}$ $\frac{A_{3.5}}{A_{2.5}} = \frac{r_{2.5}}{r_{3.5}}$ $\frac{A_{3.5}}{A_o} = \frac{2.5}{3.5}$ [M1] $A_{3.5} = 0.71 A_o$ [A1]	
4(b)(i)	Oscillations/ vibrations occur in a single direction (in a plane) that is perpendicular to the direction of transfer of energy/propagation.	B1
4(b)(ii)	Apply Malus' Law, $I_1 = I_0(\cos 40°)^2$ $I_2 = I_0(\cos 40°)^2(\cos 80°)^2$ M1 $= 0.018Io$ (2 s.f.) A1	